

Recap:

If T is a distribution associated with sequence of functions:

$$f_1(t), f_2(t), \dots$$

If $\phi(t)$ is any well-behaved (for eg. ϕ is a Schwartz).

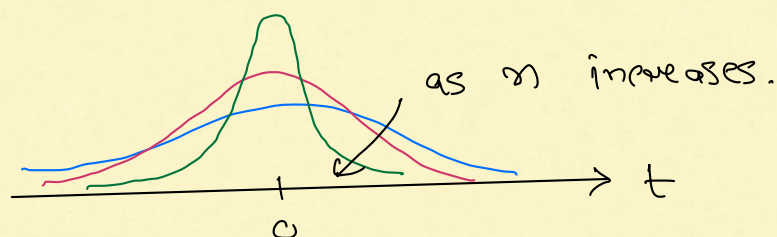
$$T * \phi : f_1 * \phi, f_2 * \phi, \dots$$

I. Examples: Convolution:

(1) $T = \delta :$

For T , we have $g_1, g_2, \dots$

$$g_n(t) = \sqrt{\frac{n}{2\pi}} \cdot \exp\left(-\frac{nt^2}{2}\right)$$



Consider $\delta * \chi :$

Analyse $g_n * \chi$ as $n \rightarrow \infty$, i.e. $\lim_{n \rightarrow \infty} \int_{-\infty}^{+\infty} g_n(t) \chi(t) dt$

$$(g_n * \chi)(t) = \int_{\tau=-\infty}^{+\infty} g_n(\tau) \chi(t-\tau) d\tau$$

→ Value of $\chi(t-\tau)$ at $t=0$ (as $n \rightarrow \infty$)
 $= \chi(t).$

$$\delta * x: g_1 * x, g_2 * x, g_3 * x, \dots$$

As $n \rightarrow \infty$ and if x is sufficiently well-behaved we have:

$$\lim_{n \rightarrow \infty} [(g_n * x)(t)] = x(t).$$

$$\Rightarrow \boxed{\delta * x = x.}$$

$\therefore$ The LTI system that acts as identity has impulse response $h(t) = \delta(t)$

$$x \rightarrow \boxed{T} \rightarrow x$$

$$x \rightarrow \boxed{\delta} \rightarrow x * \delta = x$$

2. We will consider $\delta * x$ in the integration notation:

$$(\delta * x)(t) = \int_{-\infty}^{+\infty} \delta(t) x(t-\tau) d\tau$$

= value of $x(t-\tau)$ at
time $t=0$

$$\left. \begin{array}{l} \\ \end{array} \right\} \int_{-\infty}^{+\infty} \delta(y) g(y) dy = g(0)$$

$$= x(t).$$

$$(\delta * x)(t) = \int_{-\infty}^{+\infty} \delta(t-\tau) x(\tau) d\tau$$

= $\delta(t-\tau)$ acting on $x(\tau)$ (Think of τ as the time).

using formulae from previous lectures

$$= \delta(t+\tau) \text{ acting on } x(-\tau) \rightarrow \text{time reflection}$$

New dc time shifting: $\tau \rightarrow \tau - t$:

$$\rightarrow = \delta(t + \tau - t) \text{ acting on } x(-(t - \tau))$$

$$= \delta(\tau) \text{ acting on } x(t - \tau)$$

$$= \text{value of } x(t - \tau) \text{ at } \tau = 0$$

$$= x(t).$$

$$\int_{-\infty}^{+\infty} \delta(\tau) x(t - \tau) d\tau = \int_{-\infty}^{+\infty} \delta(t - \tau) x(\tau) d\tau = x(t)$$

This is called the sifting property of δ .

("sift" $\rightarrow$ "to isolate").

$$3. \delta(t - t_0) * x(t) = x(t - t_0).$$

$\hookrightarrow$ The time-shifting property of δ .

$$x(t) \rightarrow \boxed{} \xrightarrow{\text{time-shift LTI sys.}} x(t - t_0) \quad x \rightarrow \boxed{} \xrightarrow{} x * \delta(t - t_0)$$

$\therefore \delta(t - t_0)$ is the impulse response of the time-shift LTI system.

$$4. \text{Fact: } \delta(t - t_0) * \delta(t - t_1) = \delta(t - t_0 - t_1)$$

$$\begin{array}{ccc} \delta(t - t_0) & \xrightarrow{\delta(t - t_1)} & \\ \rightarrow \boxed{} \xrightarrow{\text{shift by } t_0} \boxed{} \xrightarrow{\text{shift by } t_1} & = & \rightarrow \boxed{} \xrightarrow{\delta(t - t_0) * \delta(t - t_1)} \end{array}$$

5. Consider the LTI system:



What is the impulse response of this system?

Consider $\delta' * x$:

$$(\delta' * x)(t) = \int_{-\infty}^{+\infty} \delta'(\tau) x(t-\tau) d\tau$$

$$= \langle \delta'(\tau), x(t-\tau) \rangle, \text{ think of } \tau \text{ as time}$$

$$= - \langle \delta(\tau), \frac{d}{d\tau} x(t-\tau) \rangle$$

$$= -1 \times \text{value of } \frac{d}{d\tau} x(t-\tau) \text{ at time } \tau=0$$

$$= \left[-1 \times x'(t-\tau) \times \frac{d}{d\tau} (t-\tau) \right] \text{ at time } \tau=0$$

$$= (-1) \times x'(t-\tau) \times (-1) \text{ at time } \tau=0$$

$$= x'(t-0)$$

$$= x'(t).$$

$\therefore$ We have $\delta' * x = x'$

Hence δ' is the impulse response of this LTI system.

6. $x(t) \rightarrow \boxed{T} \rightarrow \frac{d^2 x}{dt^2}$

$$\text{Impulse response} = \delta'' \quad (= \delta' * \delta').$$

7. Periodizing.

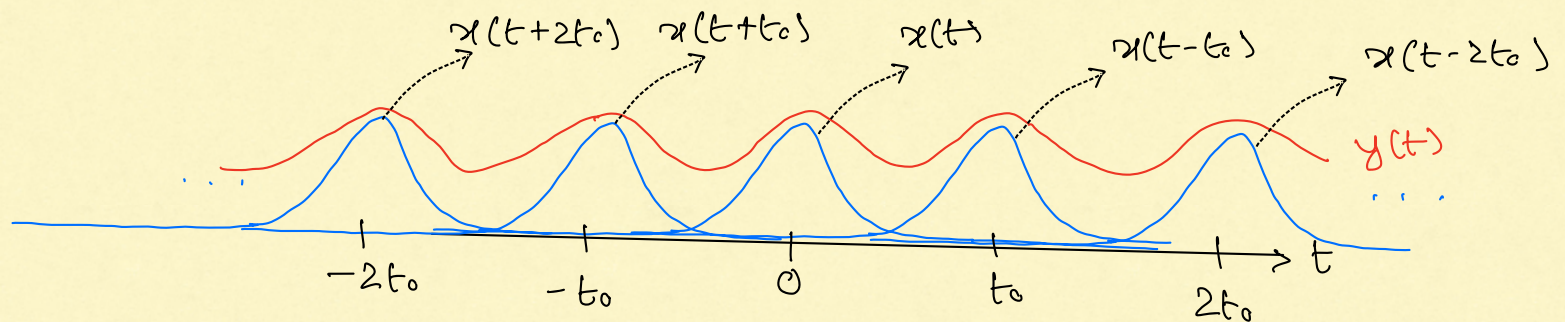
Given: a signal $x(t)$ and a constant $t_0 > 0$.

Suppose we want to obtain: $\sum_{k=-\infty}^{+\infty} x(t - kt_0) = y(t)$

This can be mathematically modelled as:

$$y(t) = x(t) * \sum_{k=-\infty}^{+\infty} \delta(t - kt_0)$$

y is periodic with period t_0 .



8. The multipath channel:

$$x \rightarrow \boxed{T} \rightarrow \sum_k a_k x(t - t_k)$$

This system has impulse response

$$h(t) = \sum_k a_k \delta(t - t_k)$$

$$x * h = h * x = \left[\sum_k a_k \delta(t - t_k) \right] * x(t)$$

$$= \sum_k a_k \cdot [\delta(t - t_k) * x(t)]$$

$$= \sum_k a_k x(t - t_k)$$

II. Meaning of Impulse Response of a system:
Example of an RC circuit.

$$x \rightarrow \boxed{T} \rightarrow x * h$$

h

$$\delta \rightarrow \boxed{T} \rightarrow \delta * h = h.$$

$$g_n \rightarrow \boxed{T} \rightarrow \approx h$$